

Year	1995	1997-1999	2000-2003	2005-2009	2010-2016	2017-2020
Major Version	Ver 0.03 to 1.09	Ver 6.x.x	Ver 7.x.x	Ver 8.x.x	Ver 9.x.x	Ver 10.x.x to 13.x.x
Key Features	Initial Release Of Postgres95 Changed to Open source use	Release of Version 6.1.1 JDBC Connector, Trigger, Constraints Multi Column Index	PL\PgSql, View Rules, PL\Python, Outer Join	Point in Time recovery, Schema, Tablespace, , column-level permissions, parallel database restore	Hot standby, 64-bit Windows, Synchronous replication, per-column collations, unlogged tables, K-nearest-neighbor indexing,	Increased robustness and performance for partitioning, transactions supported in stored procedures, parallelized vacuuming of indexes, incremental sorting

Figure 1: PostgreSQL feature additions across years and versions

level and columnar features, parallel processing to scale, and portability across operating systems.

Just to illustrate its versatility in portability, some of the operating systems supported by PostgreSQL are Linux, Windows from Win2000 onwards, FreeBSD, OpenBSD, NetBSD, Mac OS X, AIX, BSD/OS, HP-UX, IRIX, OpenIndiana, OpenSolaris, SCO OpenServer, SCO UnixWare, Solaris and Tru64 UNIX.

Postgres has some unique features:

- Non-blocking indexed or hot indexing, which allows index creation while working with tables
- Multi-version concurrency control (MVCC), enabling parallelisation of queries and taking peak advantages of cores
- Partial index, facilitating index creation on the subset of a column
- Support for storage and retrieval of a wide range of objects — for example, large objects, time range, JSONB, and XML

Notably, PostgreSQL with the PostGIS extension supports geospatial databases for geographic information systems (GIS).

Advantage 3: Versatility of Postgres creates gluiness to one platform to achieve multiple objectives

Each memory and process has a unique function. For example:

- Shared buffer helps to allocate dedicated memory to databases
 - WAL (write-ahead log) buffer allows temporary storage of changes in database
 - Work memory allows specific memory allocation per client connection
 - Maintenance memory for operations such as analyse, alter index creation
 - Writing WAL and flushing WAL data to physical files
 - Archiver that copies WAL log to physical archiver file
 - Checkpointer writing a dirty page to disk
- All these processes help the Postgres platform operate efficiently.

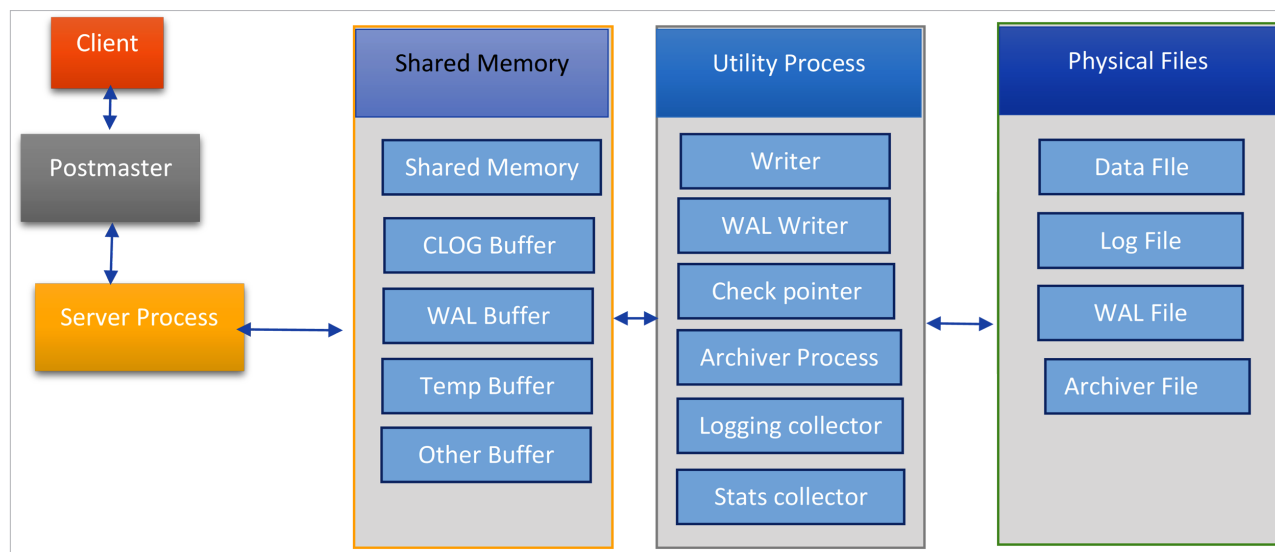


Figure 2: Postgres internal processes and architecture